

QUANTITATIVE CHEMISTRY – ATOM ECONOMY

KEY VOCABULARY

Atom economy: the % of reactant mass that becomes the useful product.

Useful product: the product you actually want to make.

By-product: a product that is not wanted.

Sustainable: uses fewer resources and makes less waste.

COMMON MISCONCEPTIONS

~~Atom economy and yield are the same.~~

✓ Atom economy is about efficiency; yield is about amount.

~~Use only the wanted product's atoms on the bottom.~~

✓ Bottom line is the total M_r of ALL products.

~~A high yield means a high atom economy.~~

✓ They are independent of each other.

ATOM ECONOMY

$$\text{atom economy} = \frac{M_r \text{ of useful product}}{M_r \text{ of all products}} \times 100$$

CONVERSION REMINDER

Higher atom economy = less waste

1 DEFINE

State what is meant by **atom economy**. [1 mark]

2 CALCULATE

$\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$. Calculate the atom economy for Cu. ($\text{Cu}=64$, $\text{H}_2\text{O}=18$) [3 marks]

Answer: _____ %

3 ONLY ONE PRODUCT

Explain why a reaction with only one product has 100% atom economy. [1 mark]

4 WHY IT MATTERS

Explain why high atom economy is good for the environment. [2 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Atom economy = (M_r of ____ product ÷ M_r of ____ products) × 100.

useful

all

less

more

6 EVALUATE

Suggest what a company could do with a valuable by-product. [1 mark]
